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NORMA ITALIAN

Gas measurement systems - Hourly gas measurement devices - Part 6: Requirements for metering units with a maximum flow rate not exceeding 10 m3 /h (size < G10)

UNI 11291-6

SEPTEMBER 2021

Gas measurement systems - Hourly based gas metering systems - Part 6: Requirements for meters with Q_{max} ÿ 10m3 /h

The rule establishes:

- which measurement data must be collected and recorded by the different devices considered;
- the minimum performance that each of the GdMs considered must have guarantee the minimum construction requirements it must meet guarantee performance;
- the appropriate methods for the remote transmission of data established measures;
- the maintenance operations necessary to ensure over time the specified performances.

The rule applies to newly built GdMs serving the Delivery points for distribution systems with meters with a maximum flow rate not exceeding 10 m3/h.

ITALIAN TEXT

This standard replaces UNI/TS 11291-6:2013.

ICS 91.140.40

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PREMISE

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CIG - Italian Gas Committee

The UNI Central Technical Commission gave its approval on July 28, 2021.

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INTRODUCTION

This standard introduces requirements relating to devices that allow the measurement of gas volumes, the compensation of gas volumes to reference conditions, the recording of measurement data, also useful for balancing distribution networks, and their remote transmission.

On October 22, 2008, the Italian Regulatory Authority for Energy, Networks and the Environment (ARERA) published Resolution ARG/gas 155/08, "Directive for the commissioning of gas metering units, characterized by minimum functional requirements and with remote reading or remote management functions, for delivery points of natural gas distribution networks." This resolution has been repealed and replaced by Resolution 631/2013/R/gas, as amended. Amendments to this regulation may be necessary as a result of measures adopted by ARERA following this resolution.

1 PURPOSE AND SCOPE

This regulation establishes:

- which measurement data must be collected and recorded by the different devices considered;
- the minimum performance that each of the GdMs considered must guarantee and the minimum construction requirements it must meet to guarantee said performance; the appropriate methods for
- remote transmission of the established measurement data; the maintenance operations necessary
- to guarantee the specified performance over time.

It applies to newly built GdMs serving the Delivery Points of distribution systems having meters with a maximum flow rate not exceeding 10 m3 /h (caliber < G10) and, in particular, it applies to GdMs of class A2 defined in UNI/ TS 11291-1.

It applies to GdMs that integrate, in a single physical module, the following functional modules: measurement, valve, volume compensation, volume recording and communication.

It defines the general requirements of the GdM and the specific requirements necessary for the use of the valve for commercial purposes pursuant to the ARERA regulation.

Note: On the date of publication of this regulation, ARERA resolution 631/2013/R/GAS is in force, which defines the following as commercial purposes in paragraph 6.4: activation or deactivation of a supply, takeover, management of arrears.

2 REGULATORY REFERENCES

This standard incorporates, by dated or undated references, provisions contained in other publications. For dated references, subsequent amendments or revisions to those publications apply only if incorporated into this standard as an update or revision. For undated references, the latest edition of the referenced publication (including any updates) applies.

UNI/TS 11291-1	Gas measurement systems - Hourly gas measurement devices - Part 1: General characteristics
UNI/TS 11291-7	of the remote management or remote reading system Gas measurement systems - Hourly gas measurement devices - Part 7: Remote management systems for gas meters - SAC, Concentrators, Repeaters and Translators Gas measurement systems - Hourly gas measurement devices - Part 10: Safety UNI/TS 11291-11-1:2014 Gas measurement systems -
UNI/TS 11291-10	Hourly gas measurement devices - Part 10: Safety
	Hourly gas - Part 11-1: General
UNI/TS 11291-11-2	Gas measurement systems - Hourly gas measurement devices - Part 11-2: Interchangeability of point-to-multipoint devices - Data model

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UNI/TS 11291-12-1:2020	Gas measurement systems - Gas measurement devices gas on an hourly basis - Interchangeability of meters with flow rate ÿ 10m3 /h with point-to-point communication - Part 12-1: General information and use cases
UNI/TS 11291-12-2	Gas Measuring Systems - Gas Measuring Devices gas on an hourly basis - Interchangeability of meters with flow rate ÿ 10m3 /h with point-to-point communication - Part 12-2: Model data
UNI EN 1359:2017	Gas Meters - Diaphragm Gas Meters
UNI EN 12405-1:2018	Gas meters - Conversion devices - Part 1: Volume Conversion
UNI EN 13611:2019	Safety and control devices for burners and appliances gas and/or liquid fuels - General requirements
UNI EN 13757-3	Communication systems for meters - Part 3: Protocols applications
UNI EN ISO 13849-1:2016	Safety of machinery - Parts of control systems safety related - Part 1: General principles for the design
IEC EN 62056-21	Electricity measurement - Data exchange for reading meters, tariff and load control - Part 21: Exchange direct data locally
IEC 61508-1:2010	Functional safety of electrical/electronic/programmable electronic safety-related systems - Part 1: General requirements

3 TERMS AND DEFINITIONS

For the purposes of this regulation, the terms and definitions set out in UNI/TS 11291-1 and the following terms and definitions.

3.1 **Remote communication:** Communication via PM1 or PP4 interface.

3.2 **local communication:** Communication via local terminal.

3.3 **operator:** Person who intervenes in the installation, maintenance and distribution exercise.

3.4 **overflow:** Condition that occurs when the “Conventional Converted Flow” reaches a value greater than or equal to 95% of the counter's Qmax .

3.5 **user:** Person who uses the gas distribution service.

3.6 **flow rate limit value:** Flow rate value, not less than Qmax, beyond which the volume measurement is no longer guaranteed.

3.7 **useful life:** Acceptable length of time for use in service. Within the scope of application of this regulation, this period is considered equal to the temporal validity of the metric stamps and of the CE marking, that is, fifteen years.

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4 TYPES OF GdM

The GdMs considered in this standard are made up of the following functional modules (figure 1) integrated into the same device:

figure 1 **Functional modules of a GdM**

- the MEASUREMENT functional module performs gas measurement using any technique (volumetric, mass, etc.);
- the VALVE module performs gas flow interception;
- the volume COMPENSATION functional module computes the volumes at reference thermodynamic conditions with at least variable *T* ;
- the DATA REGISTRATION functional module must comply with the requirements expressed from UNI/TS 11291-1 and those indicated in this standard;
- the COMMUNICATION functional module must comply with the requirements indicated by the UNI/TS 11291-1 and those indicated in this standard.

5 DATA COLLECTION AND RECORDING

5.1 General information

The GdMs covered by this standard must measure, compensate and record the volumes taking into account at least the actual temperature of the gas being measured.

In this standard all gas volumes are expressed in cubic metres and at least three decimals and, unless otherwise indicated, must be understood under the conditions of reference.

The GdMs covered by this standard which adopt the volumetric measurement of gas measure, compensate and record volumes taking into account at least the temperature according to the following formula:

$$Vb = Vm \cdot \frac{Tb}{T}$$

Where:

Vm are the measured volumes;

Vb are the volumes at reference conditions;

T is the temperature of the gas expressed in Kelvin;

Tb is the configurable reference temperature (default 288.15 K).

In case the measured value of the gas temperature is outside the working range declared by the manufacturer, the GdM must:

- generate an event (see point 7.1);
- display an “out of range” alert;
- continue to calculate the Vb considering a fixed temperature equal to Tb; the volumes measured under these abnormal conditions must be counted according to the methods for calculating volumes in the event of a metrological alarm described in point 5.4.4.

At zero flow, the GdM may not acquire the temperature measurement. At non-zero flow the acquisition of the temperature measurement, useful for the calculation of volumes, must be carried out with a periodicity no greater than 30 s.

If the flow rate limit value is exceeded, any volumes must be calculated according to the methods for calculating volumes in the event of a metrological alarm described in point 5.4.4.

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The flow rate limit value must be declared by the manufacturer.

GdMs that adopt volumetric measurement and compensation for both temperature and pressure calculate volumes at reference conditions in accordance with point 4.2 of UNI EN 12405-1:2018.

The GdMs must collect and record locally the data specified in point 5.2.

The volumes measured in each single time interval (daily/monthly) must be recorded using totalizers, carrying over the residual fractions to the next interval.

The end-of-period tariff registration must be performed by storing (in specific registers) the relevant absolute totalizers which are reset only when their maximum capacity is reached.

The various automatic controls prescribed in UNI/TS 11291-1 or referred to in this standard can detect various types of anomalies in the database; methods must be implemented that allow (e.g. SAC) to recognize whether the data was recorded under normal operating conditions of the device or not.

5.2 Essential data

The GdM must measure the gas sampled and record this measurement in at least four separate incremental totalizer registers (1 absolute totalizer register Vb Tot and 3 band totalizer registers Vb Tot F1, Vb Tot F2 , and Vb Tot F3) that cannot be modified and are reset only when their maximum value is reached. The band registers must be enabled alternately in a maximum of five daily time intervals.

In the event that, for various reasons (such as loss of time reference or unmanaged tariff program), the GdM is unable to determine the tariff band in force, the consumption must be totalized in the default band totalizer register.

The default band, unless otherwise configured, must be equal to F3.

Unless rounded off, the sum of the band totalizer registers must always correspond to the value of the absolute totalizer register.

The tariff program1) , or the timetable enabling the various totalizer registers, must have at least three coexisting types of breakdown, corresponding to: a) weekdays from Monday to Friday; b) Saturdays; c) Sundays and

holidays; and must be able to be updated up to

twice a year. The Patron

Saint's Day is broken down according

to the type of day on which it falls. The GdM must also have information indicating in which totalizer register the gas withdrawn should the time reference be lost.

The GdM must allow the configuration of at least one tariff program, which must be uniquely identified by a numeric code, as indicated by UNI/TS 11291-7.

It must be possible to configure only the future tariff program that will come into effect on a configurable date (activation date) and always at the start time of the conventional gas day.

The future tariff program, if the GdM is in “NOT CONFIGURED” mode and the activation date is in the past, must come into effect immediately.

Programming and activating a new tariff program must be recorded in the event log.

The GdM must allow the saving of absolute, band, and alarm totalizer registers in appropriate registers that cannot be modified until the next saving, with a configurable frequency, minimum every six months and maximum monthly (billing period) starting from a configurable month. Thus, once the start month and frequency are set, the saving is performed in the same months for each calendar year.

The periodicity must be configurable among the divisors of 12 (1,2,3,4,6).

1) See UNI/TS 11291-7.

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The saving of the absolute, band and alarm totalizer registers must be also carried out on the occasion of:

- entry into force of a new tariff program;
- reaching a configurable date (day, month, year) (switch date);
- valve closing, if programmed.

Saving on the asynchronous events above does not replace the periodic rescue, but adds to it and does not alter its periodicity.

5.3 Conventional flow rate calculation

The maximum conventional flow rate of the day [Qbc_max, m3 /h] and related time of occurrence is defined with the following procedure:

- the volume delivered in 15 min is recorded at reference conditions [Vb, m3];
- the average flow rate at reference conditions [Qb, m3 /h] of the period is calculated: Qb = Vb × 4;
- the calculation is repeated every 5 minutes;
- At the end of the billing period, the maximum value of Qbc_max is archived with date and time in which it occurred - hh:mm final of the 15 min of integration - (see table 1).

prospectus 1 **Example of calculation of conventional and maximum conventional flow rate**

Day 6 April 2009 Ora 08 da :mm a :mm		:05-:10	:10-:15	:15-:20	:20-:25	:25-:30	:30-:35
Volume	m3	0,320	0,024	0,041	0,068	0,080	0,300
Q 1	1,540 m3 /h	0,385					
Q 2	0,532 m3 /h		0,133				
Q 3	0,756 m3 /h			0,189			
Q 4	1,792 m3 /h				0,448		

Based on the data in Table 1, if the Q4 value remains the highest in throughout the billing period, it is assumed as the conventional flow rate value maximum and the date and time must be associated: 06/04/09 08:35.

If the overflow condition occurs, an event must be generated and must a diagnostic signal may be activated. When the overflow condition ceases, this signaling must be reset and a corresponding event must be generated.

5.4 Data presentation

5.4.1 General information

The GdM must detect and record locally in an appropriate daily data archive for at least 70 calendar days prior to the current date.

The GdM must be able to provide the necessary data so that the SAC can reconstruct the daily archive according to the scheme contained in table 2.

The GdM must be able to measure and record locally the information relating to at least twenty billing periods, to account for 13 monthly billings and 7 closings asynchronous.

The GdM must be able to provide the necessary data so that the SAC can reconstruct the archive of billing periods according to the scheme indicated in table 3.

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5.4.2 Daily Archive

The GdM must allow the SAC to reconstruct information relating to the last 70 days according to the following table:

prospectus 2 **Reconstruction scheme of data stored at the SAC level – daily data**

ID					
Data_1	In_{g_1}	(/) In	Vb (alarm)_1	(/) In	DG_1
Data_2	In_{g_2}	(/) In	Vb (alarm)_2	(/) In	DG_2
Data_3	In_{g_3}	(/) In	Vb (alarm)_3	(/) In	DG_3
Data_n	In_{g_n}	(/) In	Vb (alarm)_n	(/) In	DG_n
Data_69	In_{g_69}	(/) In	Vb (alarm)_69	(/) In	DG_69
Data_70	In_{g_70}	(/) In	Vb (alarm)_70	(/) In	DG_70

The data format must comply with the following specifications:

ID Delivery Point Identification Code (14 numeric characters).

Data_i Date of the gas day to which the data refers, expressed in day, month, year.

Vg_i Volume of gas delivered at reference conditions in the 24 hours of the gas day, with the clarification that if daylight saving time is managed, the gas volume supplied is:

- at 11pm on the gas day when we switch from solar time to daylight saving time,

- during the 25 hours of the gas day in which we return from daylight saving time to standard time.

In/- Where *V* identifies that the data, indicated in the left column, is valid; - in case contrary.

Vb (alarm)_i Volume of gas delivered, under reference conditions, in the event of an anomaly.

DG_i Daily cumulative diagnostics containing the status of the anomalies that occur they followed on the gas-day (defined by Data_i) to which the data refers. The information to be managed is reported in point 7.6.

5.4.3 Billing Period Archive

The GdM must allow the SAC to reconstruct information relating to the last 20 periods billing according to the following table:

prospectus 3 **Reconstruction scheme of data stored at SAC level – billing periods**

ID	
Id-PT N	
Data N	
Now N	
$In_{b\ Tot}$ N	In/-
$In_{b\ Tot\ F1}$ N	In/-
$In_{b\ Tot\ F2}$ N	In/-
$In_{b\ Tot\ F3}$ N	In/-
Diagnostics N	
$In_{b\ Tot\ (alarm)}$ N	In/-
Q_{bc_max} N	In/-
Date & Time Q_{bc_max} N	
Id-PT N-1	
Data N-1	
$In_{b\ Until}$ N-1	In/-
$In_{b\ Tot\ F1}$ N-1	In/-
$In_{b\ Tot\ F2}$ N-1	In/-
$In_{b\ Tot\ F2}$ N-1	In/-
N-1 Diagnostics	
$In_{b\ Total\ (alarm)}$ N-1	In/-
Q_{bc_max} N-1	In/-
Date & Time Q_{bc_max} N-1	

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Where:

ID	Delivery Point Identification Code (14 numeric characters).
Data N	Date on which billing period N was closed.
Id-PT N	Identifier of the tariff program in force in period N.
Vb Tot ^{N}	Absolute total volume stored at the end of period N.
Vb Tot Fi ^{N}	Totalized absolute volume relating to band i or the volume totalizer at stored reference conditions at the end of period N.
Diagnostics N	Cumulative diagnostics of period N, containing all the anomalies occurring from the completion of period N-1 until the closing of period N. The information to be managed is reported in point 7.6.
Vb Tot (alarm) ^{N}	Absolute totalized volume in alarm stored at closing from the N period.
Qbc_max	Maximum value of conventional flow rate occurred in period N.

N Date&Time Qbc_max N Date and time when the indicated maximum occurred.

Note The same definitions apply to period N-1.

5.4.4 Calculation of volumes in case of metrological alarm

The GdM must manage an additional absolute totalizer *(Vb* Tot (alarm)), which cannot be modified, which It only resets when it reaches its maximum value.

Volumes measured under abnormal conditions that invalidate the measurement must be counted, as well as in the absolute totalizer Vb Tot, and in the band totalizers Vb Tot F1,2,3, also in the Vb Tot totalizer (alarm).

Such abnormal conditions are, for example: exceeding the limit flow rate, measurement conditions outside the limits of correct operation of the transducers.

6 FLOW INTERCEPTION

The GdM must include a module for the automatic interception of the gas flow (of (hereinafter referred to as the shut-off valve).

The elements of the shut-off valve in contact with the gas must be constructed in compliance with the mandatory standards in function of the application and installation in places with danger of explosion.

The shut-off valve must be designed and constructed so as to ensure the correct functioning for at least 4,000 operating cycles, where each cycle is intended consisting of a closure and a reopening.

The shut-off valve shall be tested as an integral part of the GdM. The type tests to which the shut-off valve must be subjected are described in Appendix A for natural gas and in Appendix B for third family gases.

In particular, following toluene/iso-octane vapor and aging tests, must check the valve's maneuverability and that the leakage following closing is still within the prescribed limits.

The factory test shall include testing of the shut-off valve as described in Appendix C.

The shut-off valve must be able to be positioned in the “closed” state, by authorized operator, both locally and remotely.

The positioning of the shut-off valve in the “open” state must be possible only locally and with prior authorization to open.

The ability to open the shut-off valve must be transferable to the GdM from an authorized operator both remotely and locally.

The shut-off valve must not be capable of being opened remotely.

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It must be possible to open the shut-off valve on site without the use of specific equipment.

The manual opening of the shut-off valve on site must be carried out by an operator informed of the risks associated with opening with uncontrolled release of gas.

Unless otherwise programmed, in the event of any malfunction of the GdM, the shut-off valve retains the state acquired immediately before the malfunction occurred.

The detection of an anomaly in any shut-off valve actuation must not depend on the presence or absence of flow.

The actual physical "closed" or "open" position of the shut-off valve must be detectable both locally and remotely. This detection function may not be based on mechanical systems.

The closing command sent via remote or local communication must include time references that define the closing instant. The opening authorization command must include time references that define the start and end times of the authorization, respectively. Outside the time range defined by the start and end times, reopening of the shut-off valve is inhibited.

The shut-off valve can be opened using a password, in addition to local or remote control by the SAC. To this end, the GdM must provide a mechanism for an authorized operator to enter the password. The password must be securely programmable by the SAC. The password-based opening mechanism must be disabling. The password expires after its use or after a programmable number of consecutive failed attempts.

During the period in which the shut-off valve is in the closed state, the GdM activates a diagnostic function capable of detecting any residual gas flows towards the user.

If, as a result of the diagnostic function, a flow higher than that detectable by the GdM is highlighted, an alarm condition must be generated and stored in the event log and in the diagnostics2).

In the period immediately following reopening, the GdM must activate a diagnostic function capable of verifying that the opening of the shut-off valve occurred under safe conditions for the system downstream of the meter. This function must be deactivatable. To this end, after the first 10 seconds from the moment of opening, the GdM must be able to verify, within a time T ÿ 30 min, that there is no flow3) *Q* ÿ Ql where Ql ÿ 30 dm3 /h. The diagnostic function during opening must be deactivatable.

If, as a result of this diagnostic function, a flow higher than the permitted flow rate is detected, the shut-off valve must be closed immediately, an alarm condition is generated which is stored in the event log and in the diagnostics, and must be indicated on the display.

The authorization to reopen the shut-off valve remains enabled.
After three reopening attempts, the reopening authorization must be automatically cancelled and an alarm condition must be generated which is stored in the event log.

The GdM must be able to be configured to automatically close the valve at least in the following conditions:

- absence of communication for a configurable period of time; exceeding
- a threshold of break-in attempts;
- insufficient battery charge;
- unauthorized battery replacement;
- measurement system failure.

2) In addition, other functions may be activated, such as the activation of an appropriate communication extemporaneous towards the SAC.

3) For the application with third family gases, a flow rate of 12.5 dm3 /h is considered.

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The possibility that the valve is automatically closed at one or more of the above-mentioned events must be configurable.

The valve must not close, following commands, on days other than those useful4) less than explicit configuration.

It must also be documented that the safety level guaranteed by the valve is compliant as provided for by UNI EN 13611:2019 Class B or UNI EN ISO 13849-1:2016 Performance Level C or IEC 61508-1:2010 SIL 2.

7 FUNCTIONAL REQUIREMENTS

7.1 Event Log

The GdM must be equipped with an “event log” in which each event must be stored significant event, as specified below.

For each event, where this is significant, they must be identified separately.
the starting moment and the ending moment.

For each event the following must be recorded:

- The time reference of occurrence.
- The event code.
- The progressive number of the event (absolute, starting from the initialization or since the last hard reset).
- The identification code of the operator that generated the event, where applicable (see point 7.2).
- The old value, following their update, of:
  - changed parameters that influence the calculation and recording of volumes or of the courses;
  - software version;
- absolute total volume at the reference conditions at the time of the event.

The GdM must be able to record at least 180 of the above events.

Once the capacity reaches 90% of the maximum, the register must generate a specific alarm signal (see point 7.6).

Two events before the event log reaches its maximum capacity, the GdM must:

- Generate an appropriate alarm signal, separate from the previous one.
- Generate a buffer full event.
- Store each new event, always deleting the one immediately after recent preserving the buffer full event.
- Prevent changes to programming parameters that affect the calculation and recording of volumes or flow rates.
- Prevent software updates.

At least the following events must be recorded:

- clear event log;
- event buffer full;
- programming with recording of metrological and tariff parameters;
- measurements outside the metrological range;
- device failure and/or software error, specifying the functional module where possible broken down;
- clock synchronization error;

4) The “useful” day: is defined as each day of the week, delimited by the end of gas day time, other than the days indicated as holidays, including the patronal feast day of the municipality in which the return point is located, Saturdays and the days preceding Saturdays or holidays.

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- change of the operating state of the device (
- mode: “NOT CONFIGURED”, “NORMAL” and “MAINTENANCE”);
- corrupted database;
- entry into force of the tariff program;
- events related to the software update process;
- reset basic data;
- low battery;
- valve closure carried out autonomously by the GdM (following faults, anomalies or tampering attempts);
- leak detection with shut-off valve closed;
- access to electronic parts or tampering attempts (if detectable);
- unauthorized removal of a battery;
- update security credentials.

In order to avoid saturation of the event log, the following events, related to conditions which may persist or recur with close occurrence, must be recorded with criteria that avoid saturation of the register itself:

- tamper detection events;
- overflow condition detection events;
- detection of interfering magnetic field;
- leak detection with shut-off valve closed.

For example, the overflow condition detection event is stored only at the first occurrence for each gas day.

Any proprietary events defined by the manufacturer that have a repetition logic comparable, can be registered with the same criteria.

7.2 Access

Access to the GdM must be reserved for authorized persons.

To this end, both the GdM and the external systems or devices with which it should communicate any title must be able to identify themselves, through security credentials.

The GdM must be able to enumerate in a dedicated total register and non-resettable attempts to access communication and/or programming authorized.

7.3 Programming

When first switched on, the entire GdM database must assume the values of defaults specified in the relevant parts of the UNI/TS 11291-11-2 standards and UNI/TS 11291-12-2 and the operation can be limited to the totalization of the volumes under the reference conditions and communication; they may not be carried out time-based recordings until the programming is successfully concluded of the date and time. Furthermore, the GdM must be in “NOT CONFIGURED” mode.

The GdM must allow its database to be reset to defined conditions.

Following a reset, actions must be performed based on the GdM data indicated in the relevant parts of UNI/TS 11291-11-2 and UNI/TS 11291-12-2.

Once fully operational, GdM programming must be able to be carried out both locally and remotely only by operators with valid access credentials.

In full operation, remote or local programming must allow at least:

- modifying the following parameters:
  - identification code of the delivery point;
  - date and time (in the manner indicated in point 7.5);
  - parameters for volume compensation, if applicable;

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- communication parameters as defined in the relevant parts of the UNI/TS 11291-11-2 and UNI/TS 11291-12-2;
- tariff program (future only);
- default band;
- parameters for managing the billing period. Resetting the event log.

7.4 **Start-up and maintenance field operations** To highlight the start

of a configuration session, useful for example to avoid conflicts between local and remote programming, an authorised operator, be it local or remote, can put the GdM into “MAINTENANCE” mode.

In this condition, the GdM must return to “NORMAL” mode only after a specific command.

The “MAINTENANCE” mode must be detectable both locally and remotely.

The functions permitted to the maintenance operator must be configurable and must be previously authorised by the SAC.

Verification and inspection activities must be conducted in the manner and at the frequency required by current legislation.

During the verification phases, the GdM must be able to visualise the gas flow with its maximum resolution defined in the product standard5).

It must be possible to test the closing and opening of the shut-off valve.

7.5 Clock

7.5.1 General information

The GdM must base its operation on an internal clock.

The GdM clock must have a resolution of at least 1 s.

Following a power outage and restoration, the GdM checks the reliability of the clock function and, if it is unreliable, activates fault signals, logs an event, and activates a diagnostic status. Furthermore, no time-based recordings are made until the date and time have been successfully reprogrammed.

The GdM must display the data on the viewer according to local time.

The GdM must provide for the management of daylight saving time, which must be able to be deactivated both at the entrance and exit.

7.5.2 Synchronization

For the purposes of calculating volumes, the GdM must have a maximum monthly drift of the calendar clock equal to ±5 min.

Following a synchronization, the residual deviation of the clock must always be less than one minute.

Three registers R1, R2 and R3 must be managed, which cannot be reset unless the maximum capacity of the register itself is reached, of which:

R1 = register containing the number of synchronization commands accepted;

- R2 = synchronized seconds register (increasing); - R3 = synchronized

seconds register (decreasing).

R2 and R3 must be incremented simultaneously with the actual clock shift. Immediate synchronization is permitted as long as it

meets the above requirements.

5) MID Reference - MI-002 par.5.5.

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As an alternative to immediate clock synchronization, deferred synchronization is permitted, which involves a shift (measured in seconds) to be applied to the clock at defined intervals, until synchronization is achieved.

In this mode, the complete recovery of the drift to be corrected must occur in the shortest possible time, through partial movements of the clock equal to 15 s for each minute to be adjusted (unless there is a remainder less than this), which must be applied at least once per hour.

In the event that a synchronization command within the permitted limits generates a clock shift backwards whose final result is to bring the GdM back to the previous gas day, the synchronization command must be delayed to the first available instant in order to avoid the day change.

If synchronization results in a time shift greater than 4 hours, synchronization must not be performed, an event must be generated, and a diagnostic condition must be activated that could be used by the SAC to invalidate the recordings made until a new valid clock programming.

If the synchronization results in a deviation of less than 1 minute, no event shall be generated.

The clock must not undergo variations or malfunctions in the event of failure or incorrect reception of the synchronization command.

A diagnostic state must be maintained that must be activated following a synchronization command and must remain active until the synchronization process is completed.

In "NOT CONFIGURED" mode any synchronization command must be accepted and applied immediately, regardless of the limitations set in the previous points.

As an alternative to synchronization performed via explicit command, automatic synchronization is permitted, provided that it guarantees the requirements indicated above.

7.5.3 **Recordings following synchronization** If immediate

synchronization is implemented and this results in the early closure of the current period (forward synchronization), the measurement data relating to that period must still be recorded; if synchronization results in the forward movement of one or more periods relating to a tariff band, the consumption relating to the skipped periods must be considered zero.

7.6 Diagnostics and alarms

The GdM must update and make available, together with the measurement data, a register containing at least the diagnostic signals specified in Table 4.

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Table 4 **Diagnostics - Information**

Anomaly	Point	Activation	Deactivation
Lack of primary power supply	7.7	Critical battery level.	Possible battery replacement or recovery from critical battery level event.
Battery emergency	7.7	The remaining battery life is 10% of the declared battery life.	Following battery replacement or recharging.
Event Log 90%	7.1	When 90% of the total log space is exceeded Following an event events.	og reset.
Generic alarm		As long as the alarm persists.	When the alarm goes off.
Event log full	7.1	The remaining event log space is zero.	Following an event log reset.
Clock Misalignment	7.5.2	Clock synchronization would require a change of more than 4 hours. Or the GdM clock has not yet been synchronized (first start).	The clock is synchronized correctly and the variation is less than 4 hours.
Volume calculation function alarm	5.4.4	At least one anomaly was highlighted that prevents the calculation of volumes.	The volume calculation is done correctly.
Corrupted database	7.7	The database has been found to be unreliable.	When restoring the database.
Leaks present with shut-off valve closed	6	Flow detection with valve closed.	The shut-off valve is set to the open state.
Shut-off valve anomaly	6	Detection of anomalies in valve operation.	At the next movement performed with success
Tampering	7.1	Tamper detection.	Termination of the tampering condition.
Overflow	5.3	Overflow condition detected.	Overflow condition cleared.

7.7 Other requirements

7.7.1 Battery6)

For battery powered GdM, the batteries may be replaceable.

The limit operating conditions for batteries are those defined by the service levels and the reference parameters reported in the applicable standards and by the parameters defined by the references indicated in table 5.

Note For example, for meters communicating using the PP4 protocol, the boundary conditions are set out in the UNI/TS 11291-12-1 appendix A.

Table 5 **References to limit operating conditions**

Conditions Operative	PM1	PP4
Limit	Parameters of table A.2 of UNI/TS 11291-11-1:2014, also considering 4 spontaneous communications per day	Parameters of tables A.4 and A.5 of UNI/TS 11291-12-1:2020, under the conditions corresponding to one communication per day
Typical	Parameters of table A.2 of UNI/TS 11291-11-1:2014, also considering 3 spontaneous communications per day	Parameters of tables A.4 and A.5 of UNI/ TS 11291-12-1:2020, under the conditions corresponding to a communication every two days

If the GdM is equipped with only non-replaceable batteries, a battery life of of batteries equal to the useful life, when used under the limit operating conditions.

Any battery replacement must be notified in advance to the GdM and It should not take more than 255 min.

6) Please note that further fundamental requirements on the battery will be present in UNI/TS 11291-1, under revision at the date of publication of this document, and it will be necessary to refer to these requirements in any case.

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During any battery replacement or in the event of the absence of one of the replaceable batteries, the following functions must be guaranteed:

- clock;
- measurement and management of related records;
- security, as required by UNI/TS 11291-10; these functions include possible closure when battery changes are unauthorized or take too long; event recording; and persistence of stored data.

The GdM must make available information relating to the remaining autonomy time of the batteries with which it is equipped.

7.7.2 Information retention

All information recorded by the GdM must be transferred at least once each day to non-volatile memory; in the event of a power outage of any kind, the counts suspended at the time of the outage will resume from the values saved in non-volatile memory, recording the event.

The storage of programming data must follow the data retention criteria as defined in point 5.2.

The non-volatile memory shall have sufficient capacity to hold all the data required in this standard.

The GdM must provide a diagnostic function that verifies the reliability of its database. If the GdM identifies that a processed or recorded data item is unreliable, this condition must be reported so that the SAC can consider not using the data item. This means that an event must be generated and an alarm signal must be activated in the diagnostic log.

Note Alternatively, information retrieval tricks can be used, for example using different copies of the database.

7.7.3 Software Update

The GdM must provide the ability to update the software locally and remotely.

Note The update of legally relevant software modules is subject to metrological approval of the GdM.

During the communication phase related to the update of one or more program segments, all GdM functions must be maintained. Once the communication phase, which may consist of multiple steps, is complete, the GdM must verify the integrity of the received program. Only if the verification is successful can the new program be installed in place of the old one. The software update can occur extemporaneously or at a predetermined date, in this case at the conventional end-of-day time. During the activation phase, if the GdM is unable to use the rate program, it is permitted to calculate the volume in the default time slot.

If the new software update causes the historical data base to be reset, this information must be recorded in the event log.

7.7.4 Identification of the GdM

The GdM must be able to identify itself with a unique identifier generated by the manufacturer.
This identifier must contain:

- the manufacturer's identifier, as defined in EN 62056-21 (3 characters),
- an ASCII character reserved for the manufacturer,
- the measured energy: gas (coded in accordance with UNI EN 13757-3 i.e. equal to “03”) (2 characters),

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- the version (2 characters),
- the serial number (8 characters).

The manufacturer must guarantee the uniqueness of the identifier in all the devices manufactured by him. built.

The identifier must be shown in a visible and indelible manner on the plate applied to the counter also by barcode or equivalent.

8 CONSTRUCTION REQUIREMENTS

8.1 Dimensions

Dimensional requirements are specified in Appendix D.

All the components mentioned must meet the functional requirements indicated in point 7, for as applicable and with the clarifications and additions reported in the following points.

Access to batteries and electronic parts, intended to perform the functions of measurement, calculation, recording and communication, must be detected by the GdM and recorded in the event log; as an alternative to detection, the use of seals is permitted, whose break-in provides evidence of attempted entry.

8.2 Viewer

The GdM must be equipped with at least one indicator device. On this indicator a sequence described in point 8.2.1 must be available to the user and must be a service sequence is available, in which the quantities per order must be indicated point 8.2.2.

At each activation, starting from the off state of the viewer, the presentation of the information must start from the first element of the sequence dedicated to the user.

After a period of inactivity, however not less than 30 s if not more than 255 s, the display must turn off; alternatively it can remain on and return to show the first element of the sequence provided this does not cause excessive energy expenditure.

The displayed quantities must be periodically refreshed, with a period not greater than 10 s at least for fields containing dynamic information.

Leading zeros of totalizers must be displayed.

The indicating device can be of any type and technology as long as they are respected the following requirements:

- an initial test must be provided to ensure correct display of all fonts and icons;
- the size of the numeric or alphanumeric fields must be no less than 4 mm (height) and 2.4 mm (width); for sizes that require them, they must be report the units of measurement;
- readability must be guaranteed within an angle of 15° from the normal to the viewer;
- the information represented must be clearly and unambiguously highlighted; clear indications of the size can7) be used displayed or use icons and abbreviations; the size format and the abbreviations must comply with what is indicated in Table 7;
- the indicating device may be normally switched off; in this case it must be possible activate within 3 s upon specific user request;
- if the length of the information displayed exceeds the capacity of visualization, their representation must be carried out with the technique of "scroll" each character from left to right, keeping the information fixed on the display for at least 0.5 s.

7) See OIML R 137-1 and OIML R 137-2.

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In case the GdM is equipped with different indicator devices, on each device must display data relating to the functions that are implemented in the apparatus to which the device belongs in compliance with the requirements of the standards that regulate the different functions; in case the same information appears on different devices of the same GdM any possibility of misunderstanding must be avoided (which arises for example from naming different quantities with the same name) and the equality of the information in the displayed field must be ensured.

8.2.1 User Cycle Sequence

A cyclical presentation sequence must be available to the user information reported in table 6, in the order indicated.

The following information must be displayed in distinct sequences:

- software version, at least the metrological one;
- the progressive number of the reset event, if present;
- any other information, such as that relating to communication, device status and maintenance.

prospectus 6 **Viewer – Display Order**

Order	Information	Note
1	volume totalizer at reference thermodynamic conditions Vb Tot	
2	Delivery Point identification code	
3	Volume totalizer in alarm Vb Tot (alarm)	
4	identifier of the tariff program in force in the current billing period (Id-PT Current) tariff	1)
5	in force (F1/F2/F3)	1)
6	band totalizers of the current period (Vb Tot F1, Vb Tot F2,	1)
7	Vb Tot F3) diagnostics, as per table 10	
8	shut-off valve status	
9	closing date of the previous billing period	
10	totalizer of volumes at the reference thermodynamic conditions recorded at the end of the previous billing period	
11	total of alarm volumes recorded at the end of the previous billing period	
12	identifier of the tariff program in force in the previous billing period (Id-PT Precedente) band totalizers recorded at the end of the previous	2)
13	billing period	2)
14	maximum conventional flow rate in the previous billing period	
15	data	
16	no	

1) displayed on the viewer only when a tariff program is operated.
2) displayed on the viewer only when the relevant tariff program was being managed during that period.

The information reported in table 6 must be displayed in cyclic sequence, in the order indicated, by pressing a single button (dialogue button) clearly specified.

The views in the cyclic sequence must be recognizable by indicating on each display the order shown in table 6 or alternatively the relevant abbreviation reported in table 7.

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prospectus 7 **Viewer – Abbreviations**

Information	Format	Unit of measurement	Abbreviation
Data	gg-mm-aa	-	D
No	hh-mm	-	H
Current tariff	n=1,2,3		Fn
Diagnostics	2 whole digits (see table 10)		DG
Totalizer of volumes at reference conditions	5 whole numbers and three decimal places	m3	In_b or T or ÿ
Volume totalizer in alarm	5 whole numbers and three decimal places	m3	TA or ÿA
Totalizer band 1	5 whole numbers and three decimal places	m3	T1 or ÿ1
Totalizer band 2	5 whole numbers and three decimal places	m3	T2 or ÿ2
Band 3 Totalizer	5 whole numbers and three decimal places	m3	T3 or ÿ3
Delivery Point Identification Code 14 numeric characters			ID
Tariff program identifier	5 whole numbers		PT
Billing period closing date	gg-mm-aa		DF
Shut-off valve status	Coding (see table 8)		SV
Device status	0 = NORMAL mode 1 = MAINTENANCE mode 3 = NOT CONFIGURED mode		SD
Maximum conventional flow rate	2 whole digits and three decimal places	m3 /h	Q_{in}

Table 8 **Identification code for the status of the shut-off valve**

Numeric coding	Alphabetical encoding	Meaning
00	A or in clear (for example: Open)	Shut-off valve open
20 C or clear (for example: Closed)		Shut-off valve closed that cannot be enabled for reopening by entering the password
60	F or clear (for example: Error)	Shut-off valve in error
40	And or in clear (for example: Enabled) Shut-off valve enabled to reopen	
80	P or in clear (for example: PIN)	Shut-off valve closed which can be enabled for reopening by entering the password
-	A series of dashes	Valve in motion

During the presentation of the status of the shut-off valve corresponding to the code 80, long activation (ÿ 5 s) of the dialogue button, starts the procedure PIN entry; if the entered PIN is correct, the status code is set to 40.

During the presentation of the flow shut-off valve status corresponding to code 40, long activation (ÿ 5 s) of the dialogue button starts the flow shut-off valve opening procedure. When the shut-off valve interception is actually open the valve status code must assume the value 00.

When:

- the flow shut-off valve status is code 40 and
- the opening diagnostic function is enabled,

the GdM must activate the procedure for opening the valve in safe conditions (see point 6). The GdM must provide evidence of the execution of this procedure and must display the result as shown in table 9.

The GdM must provide an indication that the shut-off valve is moving, flashing the status code.

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prospectus 9 **Representation of the result of the opening procedure in safe conditions**

Sequence	Viewer indication	Occurrence	Actions
8	TEST	From the reopening of the shut-off valve and during the test	There must be a countdown timer, e.g. in 5-s steps. The indicator of this count must be placed next to the TEST indication. The countdown starts from a value consistent with the maximum test duration, which depends on the measurement technology. The dialog button during testing is inactive unless the viewer inactivity timeout expires. Only in the latter case, if the test is still in progress when the display is switched back on, the display restarts from the TEST indication.
8b	TEST - YES	At the end of the test, if the conditions for keeping the shut-off valve open have been detected	Upon successful completion of the test, activating the dialogue button restores the display of line 8 (valve status) with code equal to 00
8b	TEST - NO	At the end of the test, if the conditions for keeping the shut-off valve closed have been detected	Upon completion of the test with a failed result, activating the dialogue button restores the display of line 8 (valve status) with code equal to 40.
8c	MISTAKE	After three failed opening attempts, activation of the dialogue button restores the display of line 8 (valve status) with a code of 20 or 80 if the password is still valid. subsequent opening enablement.	

Appendix E provides a flowchart that exemplifies the behavior. described above.

8.2.1.1 Diagnostic Information Management

The diagnostic representation must be carried out according to table 10, which reports:

- The only anomalies to be displayed to the user
- Two display modes:
  - Using a numeric code.
  - Or by displaying the corresponding string.

The presence of multiple diagnostic conditions must be made clear to the user.

The display must be carried out on an anomaly-by-anomaly basis.

A 'P' must be used to indicate if the fault is still active at the time of the visualization.

The user must be able to recognize the anomaly report and cancel it by pressing a button. Cancellation must not be possible if the fault is still active.

Cancellation of anomaly reports (if not yet active) and already displayed is by pressing a button for a sufficiently long time end of display of the entire sequence of anomalies.

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table 10 **Representation of diagnostics on the display**

Anomaly (as per table 4)	Code	String
Battery emergency	01	Drums
Generic alarm	02	Alarm
Volume calculation function alarm	03	Calculation Error
Corrupted database	04	Memory Error
Leaks present with shut-off valve closed	05	Losses
Tampering	06	Tampering
Shut-off valve anomaly	07	Valve fault

In the following table 11, the diagnostic code to be displayed is shown. correspondence of the detailed diagnostic information processed by the GdM (UNI/TS 11291-11-2 and UNI/TS 11291-12-2).

Table 11 **Detailed diagnostic information and corresponding display code**

Description	Code
Critical battery level	01
Battery level less than 10%	01
General device error	02
Serious software error	02
Measurement algorithm in error	03
Memory error	04
Valve closed, but there are leaks	05
Tamper detected	06
External application interfering field	06
The valve cannot be opened or closed	07

8.2.2 Service Menu

The menu structure is defined by the manufacturer.

However, at least the current diagnostic information processed by the GdM must be made available, in hexadecimal format, in an appropriate display.

8.3 Reliability of the GdM

The manufacturer must provide evidence of the reliability of the GdM by means of an estimate of the life expected life, which must be at least equal to the useful life.

The manufacturer must indicate, providing appropriate documentary evidence, the method used for this estimate. This method must be defined by international standards that provide guidance on reliability prediction, or accelerated life testing and reliability tests and/or any other method to indicate whether the long-term stability of the GdM is affected and whether, consequently, the functions of the GdM continue to operate without loss of performance.

Note For example, for electronic instruments a predictive model is described in IEC 62059-41.

8.4 Accompanying documentation

The manufacturer must make available an instruction manual in paper or electronic format. electronic, where the manufacturer is clearly identified.

At least the following information must also be included:

- Indication of conformity to this standard.
- As necessary for the safe use of the GdM.

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- Family of gases for which the GdM is intended.
- Nominal operating conditions.
- Possible installation positions.
- Mechanical and electromagnetic environment classes.
- Safety requirements relating to commissioning and decommissioning procedures service.
- Instructions for any maintenance operations.
- Dangers arising from possible improper use.
- Criteria for verifying correct installation and operation.
- Provisions for transport and handling.
- Any indications relating to spare parts (e.g. batteries), and related requirements conservation.
- Description of any additional features.
- Type of communication protocol used, with reference to applicable standards.
- Electromagnetic compatibility and safety.

9 COMMUNICATION

9.1 Local Communication

The GdM must be able to connect locally to an external device capable of indicating and/or store at least the programming information indicated in point 7.3 and the following:

- data specified in points 5.2 and 5.3;
- remaining battery life (if applicable);
- events stored in the event log.

For this purpose, the GdM must be equipped with an optical communication port integrated and compliant, from a physical point of view, with CEI EN 62056-21.

Local connection with the external device can also occur without an internet connection physical, for example by using the same transmission medium used for communication remote and specified in point 9.2.

9.2 Remote Communication

The GdMs covered by this rule must communicate with the SAC using the PM1 profile protocols or PP4 profile protocols defined in UNI/TS 11291-1.

GdMs using the PP4 protocol must be able to be certified to the SAC through a communication forced by the installer using the external device described in point 9.1.

GdMs using the PM1 protocol must be able to be certified to the network through a communication forced by the installer using the external device described in point 9.1.

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APPENDIX A PROCEDURE FOR TYPE TESTING OF THE PERFORMANCE OF THE VALVE (regulatory) INTERCEPTION (FOR NATURAL GAS)

Step	Trial
1	The GdM shut-off valve must be placed in the open position to ensure gas flow, if it is not already open.
2	The GdM must be tested for leaks in accordance with the regulations applicable to the type of meter used (for example as provided for in point 6.2.2 of UNI EN 1359:2017).
3	If the standard applicable to the type of meter used provides for the verification of pressure absorption, the GdM must be verified in accordance with that standard (for example as provided for in point 5.2 of UNI EN 1359:2017).
4	The GdM gas outlet must be connected to a restriction that limits the gas flow to the value with 20 mbar of inlet gas pressure Q_{max}.
5	The gas pressure at the GdM inlet must be brought to 20 mbar.
6	A command (effective immediately) must be sent to the GdM to close the shut-off valve as expected under normal operating conditions. The shut-off valve must close normally and automatically.
7	The GdM outlet restriction must be removed. The gas flow through the GdM must be less than 1 dm3 /h.
8	The inlet pressure to the GdM must be increased to a value no lower than P_{max} declared by the manufacturer and 40 mbar for a period of 2 minutes. The gas flow through the GdM must be less than 1 dm3 /h.
9	The inlet pressure to the GdM must be brought to 50 mbar; the outlet to the GdM must be connected to a restriction that limits the gas flow to the value with 50 mbar of inlet pressure; Q_{max} command must be sent to the GdM to enable the opening of the shut-off valve with immediate effect; the shut-off valve must be opened following normal operating procedures; the shut-off valve must open normally, ensuring flow with the pressure absorption verified in step 3. Q_{max}

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APPENDIX B PROCEDURE FOR TYPE TESTING OF THE PERFORMANCE OF THE VALVE (normative) INTERCEPTION (FOR THIRD FAMILY GAS)

Step	Trial
1	The GdM shut-off valve must be placed in the open position to ensure gas flow, if it is not already open.
2	The GdM must be tested for leaks in accordance with the regulations applicable to the type of meter used (for example as provided for in point 6.2.2 of UNI EN 1359:2017).
3	If the standard applicable to the type of meter used provides for the verification of pressure absorption, the GdM must be verified in accordance with that standard (for example as provided for in point 5.2 of UNI EN 1359:2017).
4	The GdM gas outlet must be connected to a restriction that limits the gas flow to the value with 30 mbar of inlet gas pressure. Q_{max}
5	The gas pressure at the GdM inlet must be brought to 30 mbar.
6	A command (effective immediately) must be sent to the GdM to close the shut-off valve as expected under normal operating conditions. The shut-off valve must close normally and automatically.
7	The GdM outlet restriction must be removed. The gas flow through the GdM must be less than 0.5 dm3 /h.
8	The inlet pressure to the GdM must be increased to a value no lower than 70 P_{max} declared by the manufacturer and mbar for a period of 2 min. The gas flow through the GdM must be less than 0.25 dm3 /h.
9	The inlet pressure to the GdM must be brought to 70 mbar; the outlet to the GdM must be connected to a restriction that limits the gas flow to the value with 70 mbar of inlet pressure; a Q_{max} command must be sent to the GdM to enable the opening of the shut-off valve with immediate effect; the shut-off valve must be opened following the standard operating procedures; the shut-off valve must open normally, ensuring flow with the pressure absorption verified in step 3. Q_{max}

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APPENDIX C FACTORY VALVE PERFORMANCE TEST PROCEDURE OF INTERCEPTION

(normative)

Step	Trial
1	The inlet pressure to the GdM for natural gas applications must be brought to a value not less than 40 mbar and the gas flow through the GdM must be less than 1 dm3 /h with valve of closed interception.
2	The inlet pressure to the GdM for applications with third family gases must be brought to a value not less than 70 mbar and the gas flow through the GdM must be less than 0.5 dm3 /h with the shut-off valve closed.

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APPENDIX D DIMENSIONAL REQUIREMENTS FOR GdMs

(normative)

table D.1 **Measuring groups with G2.5 - G4 meters**

Description	Dimensions (mm)
Height (h)	ÿ 255
Length (d)	ÿ 230
Depth (p)	ÿ 185
Quota () q	ÿ 76
Wheelbase (i)	110
ISO 228-1 mount	DN 32

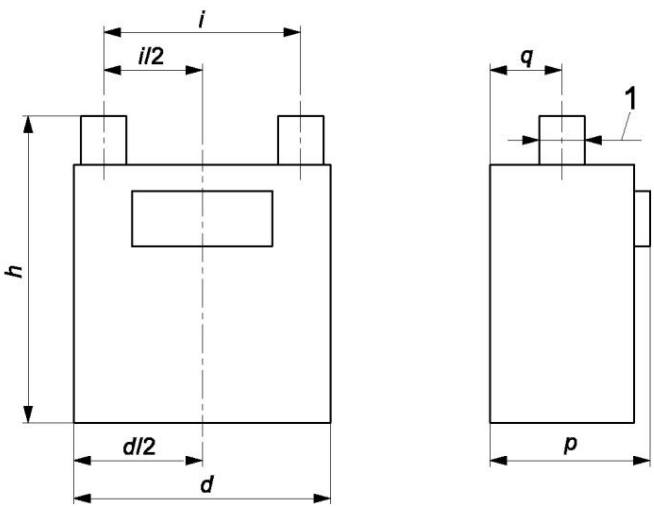
table D.2 **Measuring groups with G6 meters**

Description	Dimensions (mm)
Height (h)	ÿ 400
Length (d)	ÿ 350
Depth (p)	ÿ 226
Quota () q	ÿ 104
Wheelbase ()[i]	110 or 250
ISO 228-1 mount	DN 32

figure D.1

Legend

1 Attack



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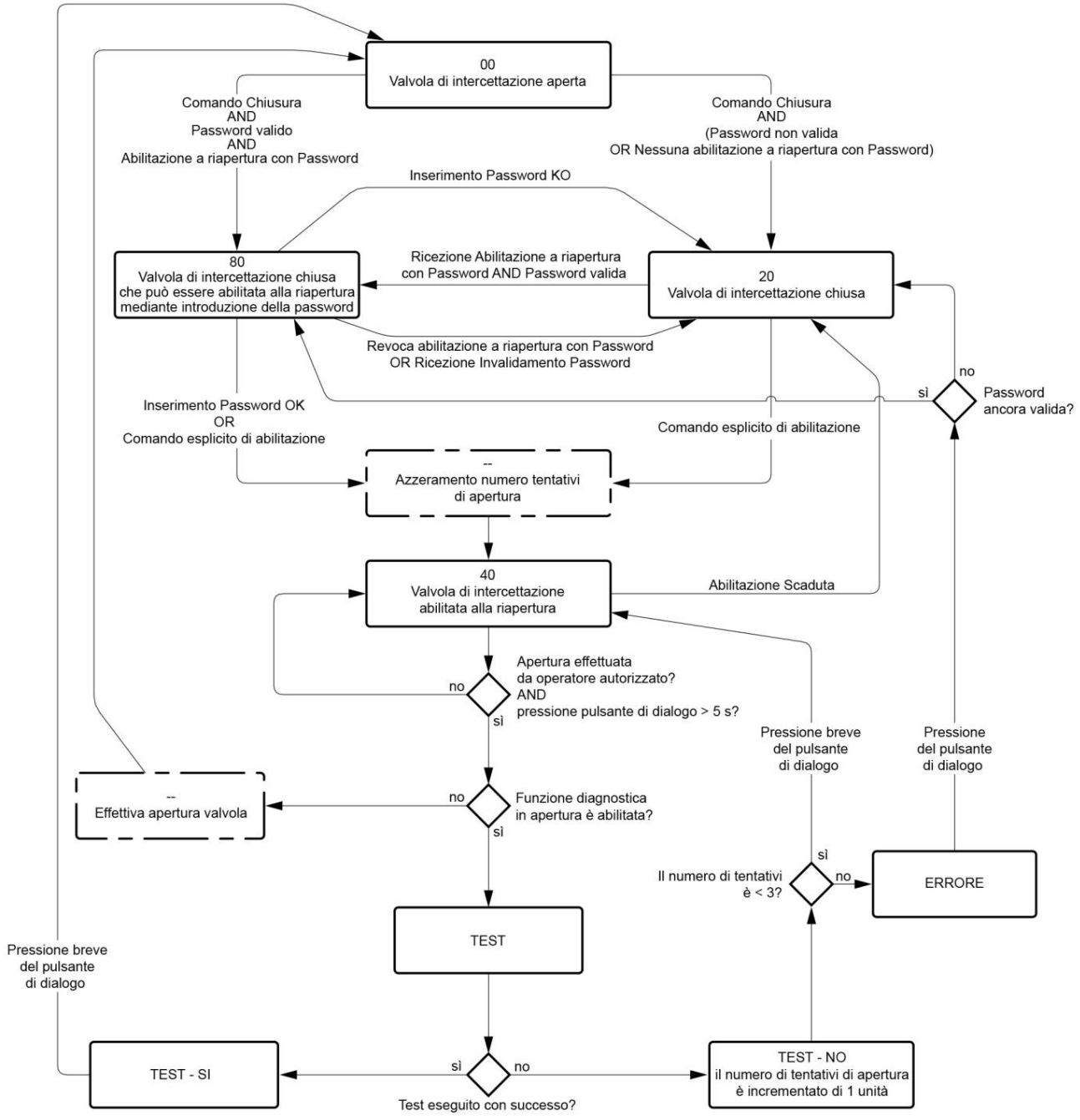
APPENDIX E MANAGEMENT OF SHUT-OFF VALVE STATUS DISPLAY

(information)

figure E.1 **State diagram of the shut-off valve**



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BIBLIOGRAPHY

Resolution ARG/gas 155/08 Directive for the commissioning of gas measuring groups, characterized by minimum functional requirements and with functions of remote reading or remote management, for the delivery points of the networks natural gas distribution

Resolution 631/2013/R/GAS Amendments and additions to the obligations for putting into service of smart meter gas

UNI EN 14236 Ultrasonic domestic gas meters

IEC EN 60870 Remote control systems and equipment

IEC EN 62059-41 Electrical Energy Measurement Devices - Reliability Part 41: Reliability Prediction

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